

The scheduler only **decides** which process should run next.
It does **not** perform the actual switch.

Responsibilities

- Select the next process.
- Apply the scheduling algorithm.
- Improve CPU utilization.
- Minimize waiting time and turnaround time.
- Maximize throughput.

Example

Ready Queue:

```
P1 → P2 → P3 → P4
```

Scheduler decisions:

- FCFS → P1
- SJF → Smallest burst time
- Priority → Highest priority
- Round Robin → First process for one time quantum

Dispatcher

Definition

The **Dispatcher** transfers CPU control to the process selected by the CPU Scheduler.
It performs the **context switch**.

Responsibilities

- Save the current process state.
- Load the next process state.
- Switch CPU execution.
- Transfer control to the selected process.

Dispatcher Latency

Dispatcher Latency is the time required to stop one process and start another.
During dispatcher latency, the CPU performs switching work instead of executing user processes.
Lower dispatcher latency results in better performance.

CPU Scheduler vs Dispatcher

CPU Scheduler	Dispatcher	Selects the next process	Executes the selection	Makes the scheduling decision	Performs the context switch	Uses scheduling algorithms	Saves and restores CPU state
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Scheduling Metrics

These metrics are used to evaluate scheduling algorithms.

1. Arrival Time (AT)

Definition

The time at which a process enters the Ready Queue.

Example

Process	Arrival Time	P1	0	P2	2	P3	5
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2. Burst Time (BT)

Definition

The total amount of CPU time required by a process to complete execution.

Burst Time includes **only CPU execution time**.

It does **not** include waiting time or I/O time.

Example

If a process needs CPU for 6 ms:

BT = 6 ms

3. Completion Time (CT)

Definition

The time at which the process completes its execution.

Example

0-----5-----9

P1 P2

Completion Times:

CT(P1) = 5
CT(P2) = 9

4. Turnaround Time (TAT)

Definition

The total time a process spends in the system.

It includes:

- Waiting time
- CPU execution time
- Any other delays until completion

Formula

TAT = CT - AT

Example

AT = 2
CT = 10

TAT = 10 - 2 = 8

5. Waiting Time (WT)

Definition

The total time a process spends waiting in the Ready Queue.

It does **not** include CPU execution time.

Formula

WT = TAT - BT

Example

```
AT = 1
BT = 4
CT = 11
```

```
TAT = 11 - 1 = 10
```

```
WT = 10 - 4 = 6
```

6. Response Time (RT)

Definition

The time from process arrival until it gets the CPU for the **first time**.

It measures system responsiveness.

Formula

```
RT = First Start Time - AT
```

Example

```
AT = 3
```

```
Starts at = 7
```

```
RT = 7 - 3 = 4
```

Important Note

For **non-preemptive scheduling**:

```
Response Time = Waiting Time
```

For **preemptive scheduling**:

```
Response Time ≠ Waiting Time
```

because a process may start execution, get interrupted, and then wait again.

What is Preemption?

Preemption means the operating system can **take the CPU away from a running process before it finishes** and give it to another process.

If preemption is **not allowed**, the running process keeps the CPU until it:

- finishes execution, or
- blocks for an I/O operation.

Non-Preemptive Scheduling

Definition

In **non-preemptive scheduling**, once a process gets the CPU, it **cannot be interrupted** by the operating system until it either:

- completes execution, or
- voluntarily gives up the CPU (e.g., requests I/O).

The scheduler waits until the running process leaves the CPU before selecting the next process.

Interview Questions

1. Why is preemptive scheduling preferred in modern operating systems?

- Better responsiveness.
 - Supports multitasking.
 - Prevents one long-running process from monopolizing the CPU.
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2. What is the disadvantage of preemptive scheduling?

- More context switches.
- Higher CPU overhead due to saving and restoring process state.

3. Which algorithms are preemptive?

- Round Robin
- SRTF
- Preemptive Priority

4. Which algorithms are non-preemptive?

- FCFS
- SJF
- Non-Preemptive Priority

avg waiting time , throughput